Online Uncertainty Quantification for the Return-to-Home Decision of a Drone: Strictly Proper Scoring, Polynomial Chaos and Kalman Filtering

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Abstract

A drone triggers its automatic return when the remaining energy no longer covers the predicted return energy plus a margin $m = \kappa E + c$. Read as a Gaussian forecast, this rule has a spread that depends on the flight only through E. We build a flight-conditional predictive law of the arrival state of charge Y. Under explicit assumptions the prediction loop has exactly five exogenous random inputs and $Y = G(\boldsymbol{\xi})$, $\boldsymbol{\xi} \sim \mathcal{N}(0, \mathrm{I}_d)$. Each source is modelled as $\mu + F\boldsymbol{\xi}$ with a germ frozen over the return, which forces the model to expose F rather than FF^{T}. Candidate models are ranked by a logarithmic CRPS skill score that splits exactly into a centre and a width term, the latter bounded by $-\log_{10}\operatorname{erf}\sqrt{\ln 2/2} \approx 0.2256$. On board, G is propagated by an unscented transform and by a Hermite polynomial chaos whose coefficients are learnt online by a Kalman filter, with a three-term bound on the decision quantile. The value of conditioning is shown to be of second order.

1. Decision problem

Let $(\Omega, \mathcal{A}, \mathbb{P})$ carry the telemetry filtration $(\mathcal{F}_t)$. The return starts at the stopping time $\tau = \inf\{t : E_{\mathrm{batt}}(t) - E(t) - m(t) \le 0\}$. A *predictive law* is an $\mathcal{F}_\tau$-measurable random probability measure $\hat{F}$ on $\mathbb{R}$; the target is $\mathcal{L}(Y \mid \mathcal{F}_\tau)$. With $q_\alpha(Y) = \inf\{t : F_Y(t) \ge \alpha\}$,

$$q_\alpha(Y) \ge s \;\Rightarrow\; \mathbb{P}(Y \ge s) \ge 1 - \alpha, \tag{1}$$

with equivalence when F_Y is not flat at level α; the converse fails for atoms. The probabilistic rule triggers when $q_\alpha(\hat{F}_t) \le s_{\min}$, i.e. uses the margin $m = \hat{y}_{\mathrm{det}} - q_\alpha(\hat{F}_t)$. The quantile is elicitable: it minimises $x \mapsto \mathbb{E}\rho_\alpha(Y - x)$, $\rho_\alpha(u) = u(\alpha - \mathbb{1}_{\{u<0\}})$ [11].

Germ representation. Tracing the firmware loop (guidance, wind model, speed map, power, current, battery) shows that uncertainty enters only through five sources: gauge error $\varepsilon_{\mathrm{SoC}}$, position error $\boldsymbol{\varepsilon}_p \in \mathbb{R}^2$, wind error $\boldsymbol{W} \in \mathbb{R}^2$, power residual δ and ground-speed error η. The Jacobian of the five loop channels with respect to them is block-diagonal with non-zero blocks, so no source is a function of the others, and omitting one lowers the Gaussian-linearised q_α by $(\sigma - \sigma')|z_\alpha| > 0$, i.e. the forecast becomes optimistic. Hence $Y = G(\boldsymbol{\xi})$, G Borel, $G \in L^2(\gamma_d)$, $d \in \{7, 11\}$.

2. Uncertainty model

Definition 1. *A model is (p, d, μ, F) with $\mu : \mathcal{S} \to \mathbb{R}^p$, $F : \mathcal{S} \to \mathbb{R}^{p\times d}$ on the state space $\mathcal{S}$; $X(s) = \mu(s) + F(s)\boldsymbol{\xi}$, so $X(s) \sim \mathcal{N}(\mu, FF^{\mathsf{T}})$; $d = 0$ is a deterministic reference.*

This is the degree-one Hermite chaos, i.e. the $L^2(\gamma_d)$ projection of any $H(\boldsymbol{\xi})$ on affine maps. The germ is drawn *once* per return: a wind or gauge error persists. Then, along states $s_1, \dots, s_N$,

$$\bar{X} = \bar{\mu} + \bar{F}\boldsymbol{\xi}, \quad \bar{F} = \tfrac{1}{N}\textstyle\sum_k F(s_k), \tag{2}$$

so $\sigma(\bar{X}) = \bar{\sigma}$ in 1-D, whereas independent steps would give $\mathrm{rms}(\sigma)/\sqrt{N}$, up to $\sqrt{N}$ too narrow. Two square roots of the same Σ_k may aggregate differently ($F_1 = \sigma\mathrm{I}$, $F_2 = -\sigma\mathrm{I}$ give $\bar{F} = 0$), so the model must expose F, taken as the unique symmetric PSD root of $\Sigma(s)$.

3. Strictly proper scoring and the CRPS

For a score $S(\hat{F}, y)$ let $S(\hat{F}, G) = \mathbb{E}_{Y\sim G} S(\hat{F}, Y)$. S is *proper* on a class $\mathcal{P}$ if $S(G, G) \le S(\hat{F}, G)$ for all $\hat{F}, G \in \mathcal{P}$, *strictly proper* if equality forces $\hat{F} = G$ [4]. Hence one criterion both ranks and calibrates.

Definition 2 ([12])**.** *On $\mathcal{P}_1$ (finite first moment),* $\mathrm{CRPS}(\hat{F}, y) = \int_{\mathbb{R}} (\hat{F}(t) - \mathbb{1}_{\{t\ge y\}})^2\, dt$.

Theorem 1. *(i)* $\mathrm{CRPS}(\hat{F}, y) = \mathbb{E}|X - y| - \frac{1}{2}\mathbb{E}|X - X'|$, $X, X' \overset{\mathrm{iid}}{\sim} \hat{F}$. *(ii)* $\mathrm{CRPS}(\hat{F}, y) = 2\int_0^1 \rho_\alpha\big(y - \hat{F}^{-1}(\alpha)\big)d\alpha$. *(iii)* $\mathbb{E}_G\,\mathrm{CRPS}(\hat{F}, Y) - \mathbb{E}_G\,\mathrm{CRPS}(G, Y) = \int (\hat{F} - G)^2$, *so the CRPS is strictly proper on $\mathcal{P}_1$.*

Proof. (iii) Expand the square: $\mathbb{E}(\hat{F}(t) - \mathbb{1}_{\{t\ge Y\}})^2 = (\hat{F} - G)^2 + G(1 - G)$ pointwise, integrate (Fubini). (i) uses $\int(\mathbb{1}_{\{t\ge a\}} - \mathbb{1}_{\{t\ge b\}})^2 dt = |a - b|$; (ii) follows from the change of variables $t = \hat{F}^{-1}(\alpha)$ [5]. □

By (ii) the CRPS averages the pinball losses of all quantiles, the decision functional of (1). Unlike the log score it is finite for point masses and 1-homogeneous; unlike the squared error it sees the spread. For $\sigma > 0$, $z = (y - \mu)/\sigma$ [8],

$$\mathrm{CRPS}(\mathcal{N}(\mu, \sigma^2), y) = \sigma\Big(z \operatorname{erf}\tfrac{z}{\sqrt{2}} + \sqrt{\tfrac{2}{\pi}} e^{-z^2/2} - \tfrac{1}{\sqrt{\pi}}\Big), \tag{3}$$

jointly convex in (μ, σ), minimised in σ at $\sigma^\star = |y - \mu|/\sqrt{\ln 2}$.

Log skill score. With a deterministic reference $\mathrm{CRPS}_{\mathrm{det}} = \sum_i |y_i - \mu_{\mathrm{det},i}|$ and marginal sum CRPS_p, let $\mathrm{crpsl} = -\log_{10}(\mathrm{CRPS}_p \,/\, \mathrm{CRPS}_{\mathrm{det}})$. $-\log$ is the only decreasing scale additive over ratios (Cauchy equation), hence antisymmetric (± 0.301 for a factor 2). With $e = \sum_i |y_i - \mu_i|$,

$$\mathrm{crpsl} = \underbrace{-\log_{10}\tfrac{e}{\mathrm{CRPS}_{\mathrm{det}}}}_{\text{centre}} + \underbrace{-\log_{10}\tfrac{\mathrm{CRPS}_p}{e}}_{\text{width}}. \tag{4}$$

Theorem 2 (width ceiling)**.** *For all μ, y with $e > 0$ and all F,* $\text{width} \le -\log_{10}\operatorname{erf}(\sqrt{\ln 2/2}) \approx 0.2256$, *with equality in 1-D iff $\sigma = \sigma^\star$.*

Proof. At $\partial_\sigma = 0$, $e^{-z^2/2} = 2^{-1/2}$, $z^2 = \ln 2$ and (3) reduces to $|y - \mu| \operatorname{erf}(z/\sqrt{2})$; sum the coordinates. □

The ceiling is the one number comparable across sources. Weights are fitted by $\min_w |\mathcal{T}|^{-1} \sum_{i\in\mathcal{T}} \mathrm{CRPS}(\mathcal{N}(\mu_i, \sigma_i(w)^2), y_i)$ on a persisted 80 % split of flights, convex when σ_i is affine in w; in-sample scores are optimistic, $\mathbb{E}J_n(\hat{w}) \le \inf J \le \mathbb{E}J(\hat{w})$.

4. Propagation

q_α of $G(\boldsymbol{\xi})$ has no closed form even for $G = \xi_1\xi_2$ (density $\pi^{-1}K_0$). Monte Carlo reaches a standard error ε only with $N = \alpha(1-\alpha)/(f(q_\alpha)^2\varepsilon^2)$ simulator calls [16], far above an on-board budget of a few calls per second.

Unscented transform [9, 17]. With $\lambda = 0$ ($\alpha_{\mathrm{UT}} = 1, \kappa = 0, \beta = 2$): $\Xi_{\pm i} = \pm\sqrt{d}\,\boldsymbol{e}_i$, $W_i^m = W_i^c = 1/(2d)$, $W_0^c = 2$. The rule integrates every polynomial of degree ≤ 3 exactly against γ_d and returns mean and covariance exactly for affine G, at $2d+1$ calls; the predictive is $\mathcal{N}(\mu_{\mathrm{UT}}, \sigma^2_{\mathrm{UT}})$.

Polynomial chaos. With normalised Hermite $h_{n+1} = (xh_n - \sqrt{n}\,h_{n-1})/\sqrt{n+1}$ and $\Psi_{\boldsymbol{a}} = \prod_j h_{a_j}(z_j)$, $\{\Psi_{\boldsymbol{a}}\}$ is a Hilbert basis of $L^2(\gamma_d)$ [2, 3, 18]: $G = \sum_{\boldsymbol{a}} c_{\boldsymbol{a}}\Psi_{\boldsymbol{a}}$, $\mathbb{E}G = c_{\boldsymbol{0}}$, $\operatorname{Var} G = \sum_{\boldsymbol{a}\neq\boldsymbol{0}} c_{\boldsymbol{a}}^2$, and the truncation residual $R_p = \sum_{|\boldsymbol{a}|_1>p} c_{\boldsymbol{a}}^2 \to 0$. Sobol indices follow from the Hoeffding decomposition [6, 15]: $S_k = V^{-1}\sum_{\operatorname{supp}\boldsymbol{a}\subseteq\mathcal{G}_k} c_{\boldsymbol{a}}^2$. The battery cut-off makes G a step: for $G = \beta\mathbb{1}_{\{z\le 0\}}$, $R_p \sim \beta^2/(\pi\sqrt{2\pi p})$ and R_2 is $1 - 2/\pi = 36\%$ of the variance. Clipping $X^+ = \max(X, 0)$ lowers the CRPS for $y \ge 0$ and gives $q_\alpha(X^+) = \max(q_\alpha(X), 0)$, so the quantile stays exact at the floor.

Theorem 3 (three-term quantile error)**.** *If G, G_p, $\hat{G}_p$ have density $\ge f_{\min}$ near q_α and $\mathcal{T}(\Delta^2) = \frac{3}{2}(2\Delta^2/f_{\min})^{1/3}$,*

$$|\hat{q}_\alpha^N - q_\alpha(G)| \le \mathcal{T}(R_p) + \mathcal{T}(\|\hat{\boldsymbol{c}} - \boldsymbol{c}\|^2) + D_N/f_{\min},$$

D_N the Kolmogorov discrepancy of the N quadrature nodes.

Proof. Triangle inequality through $q_\alpha(G_p)$, $q_\alpha(\hat{G}_p)$; Parseval turns L^2 errors into R_p and $\|\hat{\boldsymbol{c}} - \boldsymbol{c}\|^2$; a Markov bound on $\{|X - X'| > \eta\}$ optimised in η gives $\mathcal{T}$. □

The nodes are a Sobol sequence [14] mapped by $z = \sqrt{2}\,\mathrm{erf}^{-1}(2u-1)$ [13].

5. Online learning: Kalman filter on coefficients

Truncation gives the *linear* observation $G(\boldsymbol{z}) = \Psi(\boldsymbol{z})^{\mathsf{T}}\boldsymbol{c} + \varepsilon_{\mathrm{tr}}(\boldsymbol{z})$, $\operatorname{Var}\varepsilon_{\mathrm{tr}} = R_p$, $\varepsilon_{\mathrm{tr}} \perp \Psi_{\boldsymbol{a}}$. We posit the state-space model

$$\boldsymbol{c}_{i+1} = \boldsymbol{c}_i + \boldsymbol{w}_i, \quad y_{i,r} = \Psi(\boldsymbol{z}_{i,r})^{\mathsf{T}}\boldsymbol{c}_i + v_{i,r}, \qquad (5)$$

$\boldsymbol{w}_i \sim \mathcal{N}(0, Q\Delta_i\mathrm{I})$, $v \sim \mathcal{N}(0, R)$. The Kalman update [10] $\nu = y - \boldsymbol{h}^{\mathsf{T}}\hat{\boldsymbol{c}}$, $S = \boldsymbol{h}^{\mathsf{T}}\mathrm{P}\boldsymbol{h} + R$, $\hat{\boldsymbol{c}}^+ = \hat{\boldsymbol{c}} + \mathrm{P}\boldsymbol{h}\nu/S$, $\mathrm{P}^+ = (\mathrm{P}^{-1} + \boldsymbol{h}\boldsymbol{h}^{\mathsf{T}}/R)^{-1}$ is the exact Gaussian posterior. (a) $\mathbb{E}[\|\hat{G}_p - G_p\|^2_{L^2} \mid \text{data}] = \operatorname{tr}\mathrm{P}$: the posterior covariance *is* the epistemic part of the error, distinct from the aleatoric spread $\sum_{\boldsymbol{a}\neq\boldsymbol{0}}\hat{c}_{\boldsymbol{a}}^2$. (b) P^+ does not depend on y, so the optimal adaptive design equals the optimal non-adaptive one; germs are chosen greedily by the exact quantile-variance reduction $(\boldsymbol{\gamma}^{\mathsf{T}}\mathrm{P}\boldsymbol{h})^2/S$, $\boldsymbol{\gamma} = \nabla_{\boldsymbol{c}} q_\alpha = \mathbb{E}[\Psi(\boldsymbol{\xi}) \mid \Psi^{\mathsf{T}}\boldsymbol{c} = q_\alpha]$ [7]. Replacing $\sum_r \boldsymbol{h}\boldsymbol{h}^{\mathsf{T}}$ by $n\mathrm{I}$, $\mathrm{P} = (R/n)u\mathrm{I}$ with $u_{i+1} = ((u_i + \varrho)^{-1} + 1)^{-1}$, $\varrho = Q\Delta t\, n/R$: a tuning does not transfer across call rates. Normalised innovations $\nu^2/S \sim \chi^2_1$ under the model diagnose mis-tuning [1]; if the candidate pool spans H with $\dim H < P = \binom{d+p}{p}$, directions in $H^\perp$ keep their prior variance whatever the number of calls.

6. Decision criterion and value of conditioning

Survival. On a completed return with gauge reading y at arrival and reading uncertainty σ_{obs}, the true charge is modelled $\mathcal{N}(y, \sigma^2_{\mathrm{obs}})$ and the survival of a criterion $s = \hat{y}_{\mathrm{det}} - m$ is $\Phi((y-s)/\sigma_{\mathrm{obs}})$. Rules are compared at equal mean survival $\bar{S}$.

Structural bound. Let the deterministic error satisfy $\varepsilon \mid x \sim \mathcal{N}(b(x), s(x)^2)$, $\tau^2 = s^2 + \sigma^2_{\mathrm{obs}}$, so that a rule $m(x)$ has conditional mean survival $\Phi((b+m)/\tau)$. The quantile rule $m_q = -b + z\tau$, $z = \Phi^{-1}(1-\alpha)$, solves the per-flight problem $\min \mathbb{E}m$ s.t. $\Phi((b+m)/\tau) \ge 1-\alpha$ for all x. If b, τ are constant, Jensen on the concave branch of Φ shows that a constant margin is optimal for the mean-survival problem whenever $\bar{S} \ge 1/2$.

Theorem 4. *Let b, τ be independent with means $0, \bar{\tau}$ and variances v_b, v_τ. The constant margin m_c with the same mean survival as m_q satisfies*

$$m_c - \mathbb{E}m_q = \tfrac{z}{2\bar{\tau}}\big(v_b + v_\tau(z^2-2)\big) + o(v_b + v_\tau).$$

Proof. Write $u = m_c/\bar{\tau}$ and $g(b,\tau) = \Phi((b+m_c)/\tau)$. At $(0, \bar{\tau})$, $\partial_b^2 g = -u\varphi(u)/\bar{\tau}^2$ and $\partial_\tau^2 g = (2u - u^3)\varphi(u)/\bar{\tau}^2$, so $\mathbb{E}g = \Phi(u) + \frac{\varphi(u)}{2\bar{\tau}^2}\big(-uv_b + (2u-u^3)v_\tau\big) + o(\cdot)$. Equating with $\mathbb{E}\Phi((b+m_q)/\tau) = \Phi(z)$ and setting $u = z + \delta$ gives $\delta = \frac{z}{2\bar{\tau}^2}(v_b + (z^2-2)v_\tau) + o(\cdot)$; multiply by $\bar{\tau}$ and use $\mathbb{E}m_q = z\bar{\tau}$. □

The heteroscedastic term is positive only for $z > \sqrt{2}$, i.e. $\alpha < \Phi(-\sqrt{2}) \approx 0.079$, while the bias term $zv_b/(2\bar{\tau})$ is always positive but is forfeited by any predictive re-centred on $\hat{y}_{\mathrm{det}}$. Conditioning thus pays under a per-flight guarantee or with bias correction.

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